

Experiment- 5

Transfer Learning with Deep CNNs

1. Aim

Use pretrained backbones for small, specialized datasets and Fine-tune a lightweight pretrained model (VGG19/MobileNet) on Oxford Flowers with limited steps. Toggle layer freezing and plot quick accuracy/loss snapshots.

2. Theory

Introduction to Transfer Learning

In most traditional supervised machine learning approaches, it is assumed that the training data and test data are represented using the same feature space and are drawn from the same probability distribution. Under this assumption, models can perform well only when sufficient training data is available. However, in many real-world scenarios, collecting high-quality data is expensive and time-consuming, which becomes a major bottleneck in practical applications.

Semi-supervised learning attempts to reduce data requirements by using unlabelled data along with training data, but it still assumes that training and test data belong to the same domain and distribution. Active learning reduces the manual effort by selecting the most informative samples, but it is limited by the available resources and may still not provide enough data to train an accurate classifier.

Domain and Task Definitions (Formal Notation)

Domain (D)

A domain consists of:

- A feature space X
- A marginal distribution $P(x)$

So, a domain is:

$$D = \{X, P(x)\}$$

Task (T)

A task consists of:

- A label space Y
- A predictive function $f(\cdot)$

So, a task is:

$$T = \{Y, f(\cdot)\}$$

Transfer Learning Definition

Given:

- Source domain D_S and task T_S
- Target domain D_T and task T_T

Transfer learning aims to improve the target predictive function $f_T(\cdot)$ in D_T using the knowledge from D_S and T_S , where:

$$D_S \neq D_T \text{ or } T_S \neq T_T$$

This means:

- The domains may have different feature spaces or distributions
- The tasks may have different label spaces or predictive distributions

When both domains and tasks are the same, the learning becomes a traditional machine learning problem.

Homogeneous vs Heterogeneous Transfer Learning

Homogeneous Transfer Learning

Homogeneous Transfer Learning refers to a type of transfer learning in which the source domain and target domain share the same or highly similar feature representation. In this case, the input data from both domains lies in an overlapping feature space, meaning that the source and target datasets are represented in a comparable manner. This type of transfer learning is widely applied in practical scenarios where the domains involve similar kinds of data, such as transferring knowledge from one image dataset to another image dataset (image $\rightarrow$ image).

Heterogeneous Transfer Learning

Heterogeneous Transfer Learning is a more complex form of transfer learning in which the feature representation of the source domain and target domain is different. Here, the datasets are described using different feature spaces, making direct transfer difficult. Since the representations do not overlap, it becomes necessary to apply feature mapping or transformation techniques to build a relationship between the source and target domains. Due to the mismatch in feature spaces, heterogeneous transfer learning is considered more challenging compared to homogeneous transfer learning and requires additional processing steps to align or bridge the domains effectively.

Deep CNNs

Convolutional neural networks or CNNs, are a specialized kind of neural network for processing data that has a known, grid-like topology. Examples include time-series data, which can be thought of as a 1D grid taking samples at regular time intervals, and image data, which can be thought of as a 2D grid of pixels. Convolutional networks have been tremendously successful in practical applications. The name “convolutional neural network” indicates that the network employs a mathematical operation called convolution. Convolution is a specialized kind of linear operation. Convolutional networks are simply neural networks that use convolution in place of general matrix multiplication in at least one of their layers.

Deep Convolutional Neural Networks (CNNs), also called ConvNets, have become one of the most important architectures for large-scale image recognition. Their success in tasks such as image classification and object recognition was made possible due to the availability of large public image repositories like ImageNet, along with high-performance computing resources such as GPUs and distributed computing systems. A major milestone in the progress of deep CNNs was the ImageNet

Large-Scale Visual Recognition Challenge (ILSVRC), which provided a benchmark platform for evaluating and comparing image classification systems. Early approaches were based on shallow feature representations, but later deep CNN-based models achieved significantly better performance and became dominant in visual recognition tasks.

In deep learning, transfer learning is typically implemented by using a pretrained CNN backbone (trained on large datasets like ImageNet). The pretrained model learns rich hierarchical representations:

- Early layers learn general patterns (edges, textures)
- Middle layers learn shapes and structures
- Deep layers learn high-level category-specific features

For a small dataset like Oxford Flowers, transfer learning improves performance by:

- Reducing training time
- Preventing overfitting
- Reusing learned feature extraction ability

Hence, fine-tuning pretrained models such as VGG19 or MobileNetV2 is an effective strategy for building accurate classification models with limited training samples.

Merits of Transfer learning with deep CNNs

- **Reduced Training Time:**

Transfer learning reduces the training time because the model does not need to be trained from scratch.

- **Works Well on Small Datasets:**

It provides good accuracy even when the target dataset is small by using pretrained knowledge.

- **Better Feature Extraction:**

Pretrained CNNs already learn powerful features like edges, textures, and shapes, improving performance.

- **Reduced Overfitting:**

Since the model starts with pretrained weights, the chances of overfitting are reduced.

- **Faster Convergence:**

The model converges faster because weights are already optimized on large datasets like ImageNet.

Demerits of Transfer Learning with deep CNNs

- **Domain Dependency:**

If the source and target datasets are very different, transfer learning may not give good results.

- **Complexity in Layer Freezing:**

Choosing which layers to freeze or unfreeze needs careful tuning and experimentation.

- **Risk of Overfitting during Fine-tuning:**

Fine-tuning on a very small dataset can cause overfitting if not controlled properly.

- **High Computation for Large Models:**

Large models like VGG19 require high memory and more computation, making training slow.

- **Hardware Requirement:**

Deep CNN training may still require GPU support for faster training and performance.

3. Pre-Test (MCQs)

1. What is Transfer Learning?

a) Training a CNN model from scratch every time

(Incorrect because transfer learning reuses knowledge from an already trained model instead of starting from zero.)

b) Using a pretrained model for a new related task

(Correct because transfer learning applies knowledge learned from a source dataset to improve learning on a target dataset.)

c) Using only test data for training

(Incorrect because test data is not used for training in standard machine learning practice.)

d) Removing convolution layers permanently

(Incorrect because transfer learning does not require removing layers; it mainly reuses or fine-tunes them.)

Answer: b

2. Why is Transfer Learning useful for small datasets?

a) It increases the number of classes automatically

(Incorrect because transfer learning does not change the number of classes automatically.)

b) It removes the need for training

(Incorrect because training is still required at least for the final classification layers.)

c) It reduces overfitting by using pretrained features

(Correct because pretrained CNNs already contain useful feature extractors, helping the model generalize better even with less data.)

d) It converts images into text

(Incorrect because transfer learning does not perform data-type conversion.)

Answer: c

3. Which large dataset is commonly used to pretrain deep CNN models?

a) MNIST

(Incorrect because MNIST is small and does not provide rich features for large-scale object recognition.)

b) ImageNet

(Correct because ImageNet contains millions of images and is widely used to pretrain CNN backbones.)

c) Oxford Flowers

(Incorrect because it is a smaller dataset usually used as a target dataset for fine-tuning.)

d) IMDB Reviews

(Incorrect because IMDB is a text dataset, not an image dataset.)

Answer: b

4. Which of the following is a lightweight pretrained CNN model?

a) MobileNet

(Correct because MobileNet is designed for efficiency and low computational cost.)

b) VGG19

(Incorrect because VGG19 is heavy and has a large number of parameters.)

c) Inception-ResNet

(Incorrect because it is more complex and computationally expensive.)

d) AlexNet

(Incorrect because it is not as lightweight and efficient as MobileNet for modern devices.)

Answer: a

5. In Transfer Learning, “freezing layers” means:

a) Removing layers from the model

(Incorrect because freezing does not remove layers; it only stops their parameters from updating.)

b) Preventing weight updates during training

(Correct because frozen layers keep pretrained weights fixed and do not learn during fine-tuning.)

c) Increasing the learning rate of all layers

(Incorrect because freezing does not involve increasing learning rate.)

d) Training only the input layer

(Incorrect because freezing typically affects early layers, not only the input layer.)

Answer: b

6. The early layers of a CNN generally learn:

a) Very complex object-specific patterns

(Incorrect because complex patterns are learned in deeper layers.)

b) Only flower names

(Incorrect because CNN layers learn visual patterns, not text or names.)

c) Basic features like edges and textures

(Correct because early convolution layers extract low-level visual features.)

d) Only background noise

(Incorrect because CNN filters are trained to learn useful patterns, not random noise.)

Answer: c

7. Fine-tuning in Transfer Learning refers to:

a) Training the entire CNN model from scratch

(Incorrect because fine-tuning starts from pretrained weights.)

b) Retraining some pretrained layers on the target dataset

(Correct because fine-tuning updates selected layers so the model adapts to the new task.)

c) Using only validation data for training

(Incorrect because validation data is meant for evaluation, not training.)

d) Converting CNN into an SVM model

(Incorrect because fine-tuning does not change the architecture into a different algorithm.)

Answer: b

8. Transfer learning works best when:

a) Target task has limited training data and a related pretrained model is available

(Correct because pretrained weights provide strong features, helping small datasets.)

b) Target task always has unlimited data

(Incorrect because transfer learning is most useful when data is limited.)

c) Only text inputs are given for image models

(Incorrect because CNN backbones are designed for images, not raw text.)

d) No preprocessing is performed

(Incorrect because preprocessing is important for stable training.)

Answer: a

9. Which layer is commonly used to reduce spatial dimensions in CNNs?

a) Pooling layer

(Correct because pooling reduces width and height while keeping important information.)

b) Dense layer

(Incorrect because dense layers work on flattened vectors, not spatial reduction.)

c) Dropout layer

(Incorrect because dropout prevents overfitting but does not reduce spatial dimensions directly.)

d) Softmax layer

(Incorrect because Softmax is used for classification output, not dimension reduction.)

Answer: a

10. What is a major risk when transferring from an unrelated source dataset?

a) Faster training always occurs

(Incorrect because faster training is not guaranteed if the features are not useful.)

b) Negative Transfer

(Correct because transferring irrelevant knowledge can reduce target performance.)

c) Automatic dataset cleaning

(Incorrect because transfer learning does not clean the dataset.)

d) Perfect accuracy is guaranteed

(Incorrect because no method guarantees perfect accuracy.)

Answer: b

4. Procedure

Model Selection

Two pretrained deep convolutional neural network models are used for transfer learning VGG19 and MobileNetV2. Select one model.

- VGG19: Deep CNN architecture with 19 layers that uses small convolution filters to learn rich and detailed image features.
- MobileNetV2: Lightweight and efficient CNN model designed for faster training and low-resource environments. It uses depth wise separable convolutions and inverted residual blocks.

Then based on selection, Freezing/Unfreezing options:

MobileNetV2 Options:

- Base Freeze (0% unfreeze)
- 10% last layers unfreeze
- 20% last layers unfreeze

VGG19 Options:

- Base Freeze (0% unfreeze)
- 20% last layers unfreeze
- 30% last layers unfreeze

Steps

1. Start the Experiment Setup
Initialize the Python environment and import the required libraries such as TensorFlow/ Keras, NumPy, Matplotlib, and other supporting packages for training and visualization.
2. Load the Dataset
Load the Oxford Flowers dataset and organize it into training and validation sets.
Perform required preprocessing such as resizing images to a fixed input size supported by pretrained models (e.g., 224×224) and normalizing pixel values.
3. Apply Data Augmentation (If Used)
Apply image augmentation techniques such as rotation, zoom, flipping, and shifting to improve model generalization and reduce overfitting on limited data.
4. Model Selection (Choose Backbone)
Select one pretrained model backbone from the following options:
 - Option 1: VGG19
 - Option 2: MobileNetV2
 Load the selected model using pretrained weights (commonly ImageNet weights) and exclude the original top classification layers (include_top=False).
5. Build the Classification Head
Add new layers on top of the pretrained backbone to match the dataset output classes. Generally, the classification head includes:
 - Global Average Pooling / Flatten layer

- Fully Connected (Dense) layer(s)
- Dropout (optional)
- Final Dense layer with Softmax activation for multi-class classification

6. Set Freezing / Unfreezing Strategy (Interactive Toggle)

After selecting the model, choose the layer freezing option based on the model:

For MobileNetV2

- Base Freeze: Freeze all backbone layers (feature extractor only)
- 10% Unfreeze: Unfreeze only the last 10% layers of the backbone for fine-tuning
- 20% Unfreeze: Unfreeze only the last 20% layers of the backbone for fine-tuning

For VGG19

Base Freeze: Freeze all backbone layers (feature extractor only)

20% Unfreeze: Unfreeze only the last 20% layers of the backbone for fine-tuning

30% Unfreeze: Unfreeze only the last 30% layers of the backbone for fine-tuning

7. Compile the Model

Compile the model with appropriate settings such as:

- Optimizer (Adam / SGD)
- Loss function: Categorical Cross-Entropy / Sparse Categorical Cross-Entropy
- Metrics: Accuracy

Use a lower learning rate when fine-tuning to avoid disturbing pretrained weights too aggressively.

8. Train the Model with Limited Steps

Train the model for a limited number of epochs/iterations as per experiment constraints.

During training, monitor:

- Training Accuracy
- Validation Accuracy
- Training Loss
- Validation Loss

9. Plot Accuracy and Loss Snapshots

After training, plot graphs for quick comparison:

- Accuracy vs Epochs
 - Loss vs Epochs
- This provides a clear understanding of model learning behavior under different freezing/unfreezing configurations.

10. Compare Results Across Configurations

Compare all experimental results for:

- MobileNetV2 Base Freeze vs 10% vs 20% unfreeze.
- VGG19 Base Freeze vs 20% vs 30% unfreeze
Analyse which combination gives better accuracy and stable loss.

11. Final Conclusion

Conclude which pretrained backbone and which unfreeze percentage performs best for Oxford Flowers dataset, and summarize the observations based on training curves and validation performance.

5. Post-Test (MCQs)

1. What is the main purpose of unfreezing the last layers in Transfer Learning?

- a) To allow the model to adapt high-level features to the target dataset
(Correct because deeper layers capture complex task-specific patterns, so unfreezing them improves adaptation to the new dataset.)
- b) To permanently delete the pretrained backbone
(Incorrect because transfer learning uses the pretrained backbone, it does not delete it.)
- c) To stop training in the network
(Incorrect because unfreezing layers is done to enable learning, not to stop it.)
- d) To increase the dataset size automatically
(Incorrect because unfreezing layers does not increase dataset size.)

Answer: a

2. If training accuracy is very high but validation accuracy is low, it indicates:

- a) Perfect model performance
(Incorrect because perfect performance means both training and validation accuracy are high.)
- b) Underfitting
(Incorrect because underfitting happens when both training and validation accuracy are low.)
- c) Overfitting
(Correct because the model learns training patterns too well but fails to generalize.)
- d) Data preprocessing is not required
(Incorrect because preprocessing is still required for good performance.)

Answer: c

3. Why is a smaller learning rate preferred during fine-tuning?

- a) To increase memory usage
(Incorrect because memory usage is not directly controlled by learning rate.)
- b) To avoid large weight updates that can damage pretrained features
(Correct because fine-tuning requires small updates to preserve useful pretrained weights.)
- c) To make the dataset larger
(Incorrect because learning rate does not affect dataset size.)
- d) To remove the need for backpropagation
(Incorrect because training always uses backpropagation.)

Answer: b

4. In MobileNetV2, base freeze means:

- a) Training the entire model from scratch
(Incorrect because base freeze does not train the whole backbone again.)

- b) Freezing all backbone layers and training only the added classifier layers
(Correct because the backbone works as a fixed feature extractor and only the new layers learn.)
- c) Removing the classifier layers
(Incorrect because classifier layers are required for prediction.)
- d) Increasing the image size automatically
(Incorrect because image resizing is done in preprocessing.)

Answer: b

5. Unfreezing 20% of the last layers mainly helps in:

- a) Learning only basic edges and textures
(Incorrect because basic features are learned in early layers.)
- b) Increasing batch size automatically
(Incorrect because unfreezing layers does not change batch size.)
- c) Learning task-specific patterns for the target dataset
(Correct because deeper layers learn complex features that are more target-dependent.)
- d) Converting the CNN into an SVM model
(Incorrect because unfreezing does not change the algorithm type.)

Answer: c

6. Which pretrained model is generally more computationally heavy?

- a) VGG19
(Correct because VGG19 has many parameters and requires higher computation.)
- b) MobileNetV2
(Incorrect because it is designed to be lightweight and efficient.)
- c) KNN
(Incorrect because KNN is not a deep CNN model.)
- d) Naive Bayes
(Incorrect because it is not a CNN architecture.)

Answer: a

7. Which plot best helps in checking training performance and overfitting?

- a) Histogram of file names
(Incorrect because file names do not show training behavior.)
- b) Scatter plot of random points
(Incorrect because random scatter plots do not reflect model learning.)
- c) Confusion matrix before training
(Incorrect because confusion matrix is useful after predictions, not before training.)
- d) Accuracy and loss curves
(Correct because these curves show learning progress and generalization issues.)

Answer: d

8. If validation loss increases while training loss decreases, it usually means:

- a) The model is underfitting
(Incorrect because underfitting means training loss is also high.)
- b) The model is generalizing perfectly
(Incorrect because increasing validation loss indicates poor generalization.)
- c) The model is overfitting
(Correct because the model fits training data well but performs worse on unseen validation data.)
- d) The dataset has no images

(Incorrect because training cannot proceed properly without images.)

Answer: c

9. What is the purpose of the Softmax layer in CNN classification?

a) To reduce image size

(Incorrect because pooling is mainly used for reducing spatial size.)

b) To normalize dataset images

(Incorrect because normalization is a preprocessing step, not Softmax.)

c) To detect edges in images

(Incorrect because convolution layers learn edge detectors.)

d) To convert outputs into probabilities over classes

(Correct because Softmax outputs a probability distribution across multiple classes.)

Answer: d

10. Negative Transfer occurs when:

a) Transfer improves accuracy strongly

(Incorrect because this is positive transfer.)

b) Only classifier layer is trained

(Incorrect because this does not necessarily cause negative transfer.)

c) Learning rate is small

(Incorrect because small learning rate usually stabilizes training.)

d) Transfer from an unrelated source reduces target task performance

(Correct because irrelevant transferred knowledge harms target learning.)

Answer: d

6. References

1. I. Goodfellow, Y. Bengio, and A. Courville, "Deep Learning," MIT Press, Cambridge, MA, USA, 2016.
2. Q. Yang, Y. Zhang, W. Dai, and S. J. Pan, Transfer Learning. Cambridge: Cambridge University Press, 2020.
3. K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," Apr. 2015.